CS6350 — Computer Vision

Tier	B
Phase (order)	7
Do this first	CS6910, linear algebra
Estimated effort	12 weeks
Unlocks	CS6790, CS6777

Roadmap

1. Image formation, filtering, features (SIFT/Harris).
2. Geometry (cameras, epipolar), then CNN-based detection and segmentation.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Stanford CS231N Deep Learning for Computer Vision I 2025 (18 videos). Stanford CS231N *Deep Learning for Computer Vision* 2025 (official Stanford Online).

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- First Principles of Computer Vision: Introduction (Shree Nayar) (7 videos). Shree Nayar *First Principles of Computer Vision* (Columbia): classical vision (image formation, features, stereo, flow). Other units are separate playlists on the same channel.
- Deep Learning for Computer Vision (22 videos). Michigan EECS 498 *Deep Learning for Computer Vision* (Justin Johnson): CS231n's successor, with assignments.
- Deep Learning for Computer Vision (100+ videos). NPTEL *Deep Learning for Computer Vision* (Vineeth Balasubramanian, IIT Hyderabad).
- Computer Vision and Image Processing - Fundamentals and Applications (42 videos). NPTEL *Computer Vision and Image Processing* (IIT Guwahati): classical image processing.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–3	Image classification, linear classifiers, regularisation, optimisation. Assignment 1 (kNN, SVM, softmax) from cs231n.github.io.
2	4–6	Backprop, CNNs, CNN architectures (ResNet etc.). Assignment 2: CNN on CIFAR-10.

Week	Lectures	What to do
3	7–8	RNNs, attention, transformers.
4	9–10	Detection, segmentation, visualisation, video. Fine-tune a YOLO / Mask R-CNN model.
5	11–14	Distributed training, self-supervised learning, generative models (VAE, GAN, diffusion).
6	15–18	3D vision, vision-language, robot learning, human-centred AI. For classical CV (filters, geometry, stereo) use Shree Nayar's series (alt).

▼ **All lectures in *Stanford CS231N Deep Learning for Computer Vision I 2025* (18 videos)**

1. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 1: Introduction
2. Stanford CS231N | Spring 2025 | Lecture 2: Image Classification with Linear Classifiers
3. Stanford CS231N | Spring 2025 | Lecture 3: Regularization and Optimization
4. Stanford CS231N | Spring 2025 | Lecture 4: Neural Networks and Backpropagation
5. Stanford CS231N | Spring 2025 | Lecture 5: Image Classification with CNNs
6. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 6: CNN Architectures
7. Stanford CS231N | Spring 2025 | Lecture 7: Recurrent Neural Networks
8. Stanford CS231N | Spring 2025 | Lecture 8: Attention and Transformers
9. Stanford CS231N | Spring 2025 | Lecture 9: Object Detection, Image Segmentation, Visualizing
10. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 10: Video Understanding
11. Stanford CS231N | Spring 2025 | Lecture 11: Large Scale Distributed Training
12. Stanford CS231N | Spring 2025 | Lecture 12: Self-Supervised Learning
13. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 13: Generative Models 1
14. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 14: Generative Models 2
15. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 15: 3D Vision
16. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 16: Vision and Language
17. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 17: Robot Learning
18. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 18: Human-Centered AI

Books, notes and websites

- Szeliski, *Computer Vision: Algorithms and Applications* — szeliski.org/Book (free)
- Stanford CS231n; OpenCV tutorials

Practice / proof of understanding

Build a panorama stitcher with OpenCV and train a small object detector.

What your IITM course sheet says

From the PDF: CS6350

Official description: Computer Vision focuses on development of algorithms and techniques to analyze and interpret the visible world around us. This requires understanding of the fundamental concepts related to multi-dimensional signal processing, feature extraction, pattern analysis visual geometric modeling, stochastic optimization etc. Knowledge of these concepts is necessary in this field, to explore and contribute to research and further developments in the field of computer vision. Applications range from Biometrics, Medical diagnosis, document processing, mining of visual content, to surveillance, advanced rendering etc.

Official syllabus (trimmed):

> Syllabus: • Digital Image Formation and low-level Processing: Overview and State-of-the-art, Fundamentals of Image Formation, Transformation: Orthogonal, Euclidean, Affine, Projective, etc; Fourier Transform, Convolution and Filtering, Image Enhancement, Restoration, Histogram Processing. • Depth estimation and Multi-camera views: Perspective, Binocular Stereopsis: Camera and Epipolar Geometry; Homography, Rectification, DLT, RANSAC, 3-D reconstruction framework; Auto-calibration. • Feature Extraction: Edges – Canny, LOG, DOG; Line detectors (Hough Transform), Corners – Harris and Hessian Affine, Orientation Histogram, SIFT, SURF, HOG, GLOH, Scale-Space Analysis- Image Pyramids and Gaussian derivative filters, Gabor Filters and DWT. • Image Segmentation: Region Growing, Edge Based approaches to segmentation, Graph-Cut, Mean-Shift, MRFs, Texture Segmentation; Object detection. • Pattern Analysis: Clustering: K-Means, K-Medoids, Mixture of Gaussians, Classification: Discriminant Function, Supervised, Un-supervised, Semi-supervised; Classifiers: Bayes, KNN, ANN models; Dimensionality Reduction: PCA, LDA, ICA; Non-parametric methods. • Motion Analysis: Background Subtraction and Modeling, Optical Flow, KLT, Spatio-Temporal Analysis, Dynamic Stereo; Motion parameter estimation. • Object Recognition: Structural, model-based, appearance and shape-based methods; probabi

Official textbooks: 1. Richard Szeliski, Computer Vision: Algorithms and Applications, Springer-Verlag Limited, London, 2011.

Official references: 1. Richard Hartley and Andrew Zisserman, Multiple View Geometry in Computer Vision, Second Edition, Cambridge University Press, March 2004. 2. K. Fukunaga; Introduction to Statistical Pattern Recognition, Second Edition, Academic Press, Morgan Kaufmann, 1990. 3. R.C. Gonzalez and R.E. Woods, Digital Image Processing, Addison-Wesley, 1992. 4. IEEE Transactions on Pattern Analysis and Machine I